Cancelling Translated Copies of Points in Convex Position

Problem Statement

Definition 1. *A finite set S of points in the plane is in* convex position *if every point of S is a vertex of its convex hull.*

Definition 2 (Setup)**.** *Let S be a finite set of $n \geq 1$ points in $\mathbb{R}^2$ in convex position. Let T be a finite* set *of translating vectors in $\mathbb{R}^2$ (each vector appears at most once), each assigned a sign $\sigma(t) \in \{+1, -1\}$.*

For every point $p \in \mathbb{R}^2$, define

$$\sigma(p) \;=\; \sum_{\substack{t \in T \\ \exists\, s \in S:\, p = t+s}} \sigma(t).$$

Theorem 1 (Lower Bound – Known)**.** *Under the above setup, there are at least n points $p \in \mathbb{R}^2$ for which $\sigma(p) \neq 0$.*

Conjecture 1 (Main – To Be Proved)**.** *Assume:*

- $|S| \geq 3$ *(at least three points in convex position), and*
- $|T| \geq 2$*, with T containing at least one vector with $\sigma(t) = +1$ and at least one with $\sigma(t) = -1$.*

If $|\{p \in \mathbb{R}^2 : \sigma(p) \neq 0\}| = n$, then $n = 4$ and S is the set of vertices of a parallelogram.

Remark 1 (Nontriviality)**.** *The hypothesis $|T| \geq 2$ with both signs is necessary: a single translate gives $|supp(\sigma)| = n$ for any S. The hypothesis $|S| \geq 3$ excludes degenerate one- and two-point sets. The conjecture says: once genuine cancellation occurs (both signs present) but equality $|supp(\sigma)| = n$ still holds, S must be a parallelogram — and in particular $n = 4$. For $n = 3$ the equality condition turns out to be unachievable, so the conjecture holds vacuously in that case (proving this is part of the proof).*

Goal

Prove Conjecture **??** from scratch. Concretely:

1. Re-prove (or cite) the lower bound $|\mathrm{supp}(\sigma)| \geq n$ via the Lonely Points Lemma.
2. Show that if T has both signs and $|\mathrm{supp}(\sigma)| = n$, then $n \leq 4$.
3. Show that $n < 4$ is impossible under equality, so $n = 4$.
4. Show that a convex quadrilateral achieving equality must be a parallelogram.

Remark 2. *The lower bound follows from the Lonely Points Lemma (Blokhuis–Chen 2018): for each $s \in S$ there is a* lonely *point $\ell_s = s + t_s \in S + T$ whose unique preimage is s, hence $\sigma(\ell_s) = \sigma(t_s) \neq 0$; these n points are distinct. The content of the conjecture is that the extremal case $|supp(\sigma)| = n$ forces S to be a parallelogram.*